

Chapter 5 — Intrinsic failure

What breaks with nobody attacking you

No adversary in this chapter. Everything here happens when the system is working exactly as designed.



Hallucination is a mechanism, not a mystery

- Chapter 1: the objective is to predict a plausible next token. Nothing in that objective rewards *knowing* over *sounding right*.
- Where the training data was dense, plausible and true coincide. Where it was thin, they come apart — and nothing in the system detects which regime it is in.
- **So the failure is not a bug in an otherwise truthful system. Truthfulness was never the objective.**

Thread — truthfulness. In Chapter 2 you rated it as a rubric dimension. Here it becomes a structural property. Chapter 14 turns it into a work habit.



The band it lives in

- Common knowledge — reliable. There is too much agreeing evidence to go wrong.
- Genuinely obscure — usually declined. The model has nothing and often says so.
- The band between — believes it knows, and confabulates fluently.

Your own literature sits in that band. Not obscure enough to be declined, not common enough to be safe. That is why your experience is that the tool is reliable on textbook material and erratic on your research, and it is not random.



The default academic failure

- A fabricated citation with plausible authors, a plausible year and a plausible venue.
- It is not caught by reading the sentence. It is caught by **resolving the identifier**.
- Worse: the near-miss. A real paper, correctly cited, that does not say what it is cited for.

TODO(produce) — the failure gallery: ten real wrong outputs from mechanics, concrete and fluids. Each carries the prompt, the output, what is wrong, and **how it was caught**. The last field is the one that teaches.



Calibration

- Stated confidence is generated by the same process as the answer. It is a token sequence, not a measurement.
- "I'm certain" and "I think" are style, and both were rated during training.
- **Confidence and correctness come apart** — and you have already produced that data yourself, in Chapter 2, on your own tally sheet.



Arithmetic, units and significant figures

- Arithmetic is done by prediction, not by calculation. It is right most of the time, which is worse than being wrong reliably.
- Significant figures are copied from the surrounding style, not derived from input precision. A value quoted to six figures from two-figure inputs looks authoritative and is not.
- Units are the failure that survives review, because the number still looks like a number.

TODO(capture) — demonstrate on a coefficient from mechanics, concrete or fluids where the group knows the right answer cold. The point lands only if the room can see the error without being told.



Why benchmark scores will not tell you

- A benchmark measures a distribution of tasks. Yours is one point, and it is not in that distribution.
- Contamination is real and rarely quantified: benchmark items appear in training data.
- A model that improves on a leaderboard may be worse on your problem, and no published number will tell you.

Which is Chapter 4's argument arriving from the other direction. **Your five cases beat their thousand.**



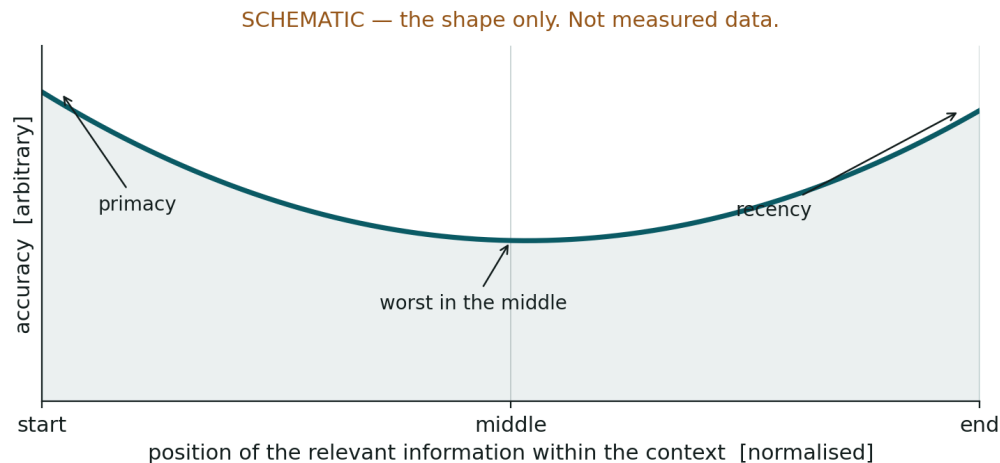
Long context: cost has an equation, accuracy does not

Chapter 7 derives what long context **costs**. There is no equivalent derivation for what it does to **accuracy**. There are empirical curves and mechanistic arguments, and that is all there is.

- Saying so is not a gap in the course. It is the finding.
- A room that has been handed a clean equation for one and an honest "we have curves" for the other will remember which is which.



The position curve



- Accuracy depends on **where** the relevant information sits, not only on how much context there is.
- Best at the beginning and the end. Worst in the middle.
- The measured version covers two tasks — multi-document QA and synthetic key-value retrieval — under greedy decoding, and **one model tested was near-perfect on the synthetic task**. The effect is real and it is not universal.



Reproducing that figure properly

The measured curves may be reproduced in this deck. The licensing detail decides how.

- The TACL version of record is CC BY 4.0 — reproducible with attribution.
- The arXiv version is not. It carries a non-exclusive distribution licence that grants no redistribution right.
Same paper, different rights.
- The authors' code is MIT, so the curve can be regenerated in house style instead of copied.

Two versions of one paper with different reuse rights is exactly the trap Chapter 18 asks you to notice. The figure above is a schematic and says so on its face.



Needle in a haystack

- Insert one distinctive sentence at a known depth, ask for it back, sweep depth against context length, and plot the grid.
- **There is no paper.** It is a widely used open-source implementation, and a heatmap from it is an implementation output, not a published result. Label it as such.
- It tests **literal matching**. A model can pass it and still fail when the question shares no words with the answer.



Context rot, and who is telling you

- Extended needle variants across 18 models report degradation rising with input length even on tasks the simple probe calls solved.
- The authors state they have **no mechanistic explanation**, and that their tasks do not cover real-world synthesis — where they *expect* worse. That expectation is stated, not measured.

Conflict of interest. The organisation publishing this sells retrieval infrastructure. "Long context degrades" is commercially convenient for them. The work looks careful; the interest is declared here because that is the standard this course asks of you.



The nearest thing to a proof

Softmax attention coefficients must **disperse** as the number of items grows. With a fixed logit scale, the operation cannot sharply select from an arbitrarily large set.

- That is a real result, proved, on a task as simple as finding the maximum key.
- **It bounds sharpness of selection. It does not bound task accuracy.** Those are different quantities and the gap between them is not bridged.
- Demonstrated on small controlled tasks, not production benchmarks.

Do not let this be quoted as "proof that long context degrades". It is the strongest mechanistic argument available, and it is still not that claim.



Supporting mechanisms — none of them proofs

- Positional encoding extrapolating beyond the lengths trained on.
- The long-context regime is undertrained — few training sequences reach the maximum length, so the far end of the window is the least practised.
- **Nominal against effective context.** A model advertising a large window may hold accuracy over far less of it, and benchmarks built to test this report exactly that.

Each is plausible and partially evidenced. Presented together they feel like an explanation. They are a list of candidates, and saying so is the honest version.



Model drift

- A working prompt has a shelf life. The model behind it changes and yours did not.
- Pin versions where you can. Log the model ID and date beside every result. Re-test before relying on it again.
- Small segment, highest return per minute in the course.

This is why Chapter 4's worksheet has a header block, and why Chapter 18 makes the log a governance requirement rather than a good habit.



AI-text detection — and it is not watermarking

- **Detection** infers origin from the text. **Watermarking** embeds a signal at generation. Different mechanisms, different reliability, different people affected.
- Detectors are unreliable in **both** directions.
- Seven detectors tested over 91 human-written TOEFL essays returned a mean false-positive rate of **61%** — against near-zero on US 8th-grade essays. One rewording prompt cut it to 12%.

For an international research group this is the practical finding of the chapter. The direction is well supported. The specific percentages are a March 2023 snapshot of products that have since changed — quote the direction, date the numbers.



What to do if you are accused

- The accusation is unfalsifiable by construction. You cannot prove you wrote something.
 - **What protects you is process, not proof:** version history, drafts, notes, and a disclosure you wrote before anyone asked.
 - Ask what instrument was used and what its false-positive rate is on writers like you. Most accusers will not know.
-

Chapter 17 meets this again from the other side, and Chapter 18 turns it into a lab standard.



Watermarking is a different topic

- A published scheme modifies the sampling procedure so generated text carries a detectable signal, evaluated at scale in production.
- It is **text only**. Do not let it be cited for images or audio; those are separate systems.
- And it tells you about text from *that* system. It says nothing about text from anywhere else.



"LLMs are black boxes" — the correction

That framing is too strong, and by now it is lazy. The accurate word is **partially instrumented**.

- **Sparse autoencoders** decompose activations into interpretable features — first on a one-layer model, later at production scale.
- **Attribution graphs** trace the computational path for a specific prompt.
- These are real tools, they cost real money to run, and they are themselves under validation.



Four qualifications, on the same slide as the claim

1. It traces features and computational paths, not "which weights were used".
2. Coverage is partial. The authors report satisfying insight for *about a quarter* of the prompts they tried.
3. The bottleneck is human interpretive labour as much as compute.
4. Faithfulness is open — in the authors' words, they "lack rigorous methods for evaluating whether our features faithfully capture model computations".

Conflict of interest. This work is published by the lab that makes the models it interprets. A favourable interpretability result is commercially useful to them. The qualifications above are the authors' own, which is a point in their favour — and the interest is still declared.



The distinction people miss

The model's stated reasoning is **not** an explanation.

- Inject a bias into the prompt — reorder the options so the answer is always the first one — and models switch answers and **rationalise, without ever mentioning the bias**. Accuracy fell by up to 36%.
- On modern reasoning models the same holds: the chain of thought reveals the hint it actually used **often below 20% of the time**.
- The authors narrow it themselves, and the narrowing matters: their hints are easy to exploit, so this shows chain-of-thought monitoring is **unreliable**, not that reasoning is always post-hoc.

"It told me why" is not "I know why."



Where this leaves us

- Hallucination as a property of the objective, living in the band your own literature occupies.
- Accuracy over long context: curves and mechanisms, one real proof that does not prove the thing people quote it for.
- Detection that fails hardest on the people in this room.
- A system that is partially instrumented — and stated reasons that are not causes.

Nobody has attacked you yet. Chapter 6 is what happens when someone does.

